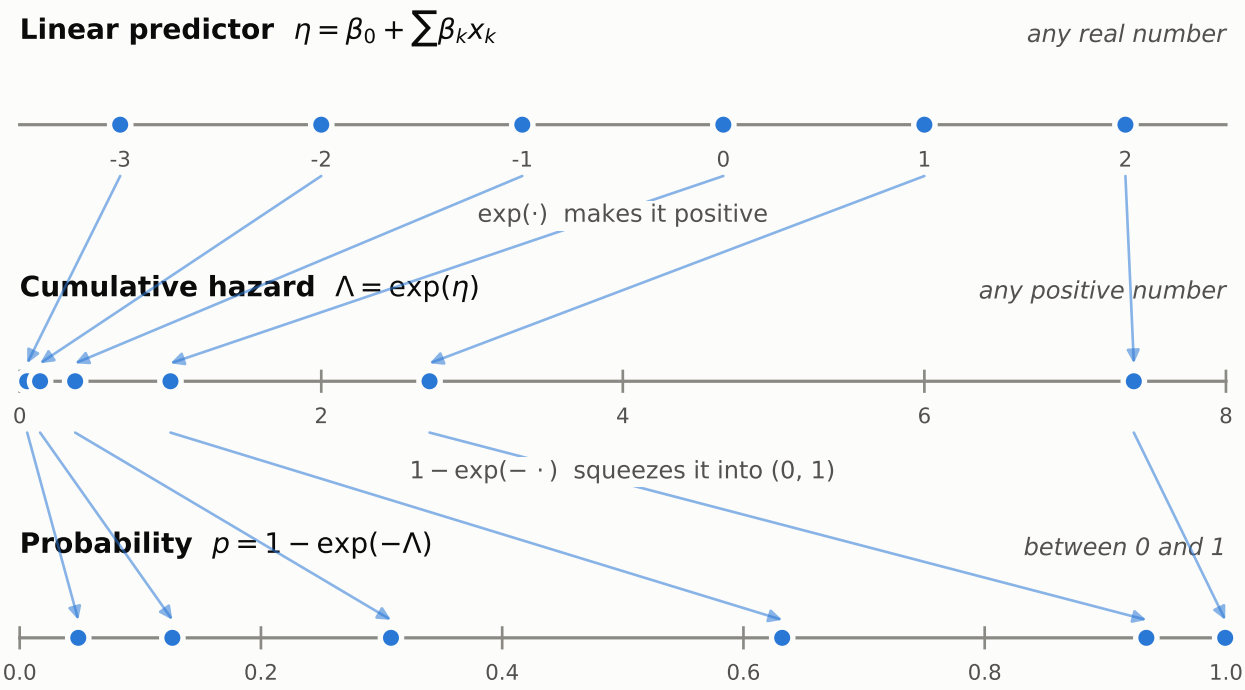
THE SHORT ANSWER

Is the probability the final scale?

Yes. The probability p is what the model finally outputs: the chance that a unit has an event within its window. It is the number that is reported, and the one that can be compared with what actually happened (calibration).

But the model is not built on the probability scale. It is built on the linear predictor η, a weighted sum of the covariates, and two fixed steps then carry η to p. The figure follows six values of η through both steps.

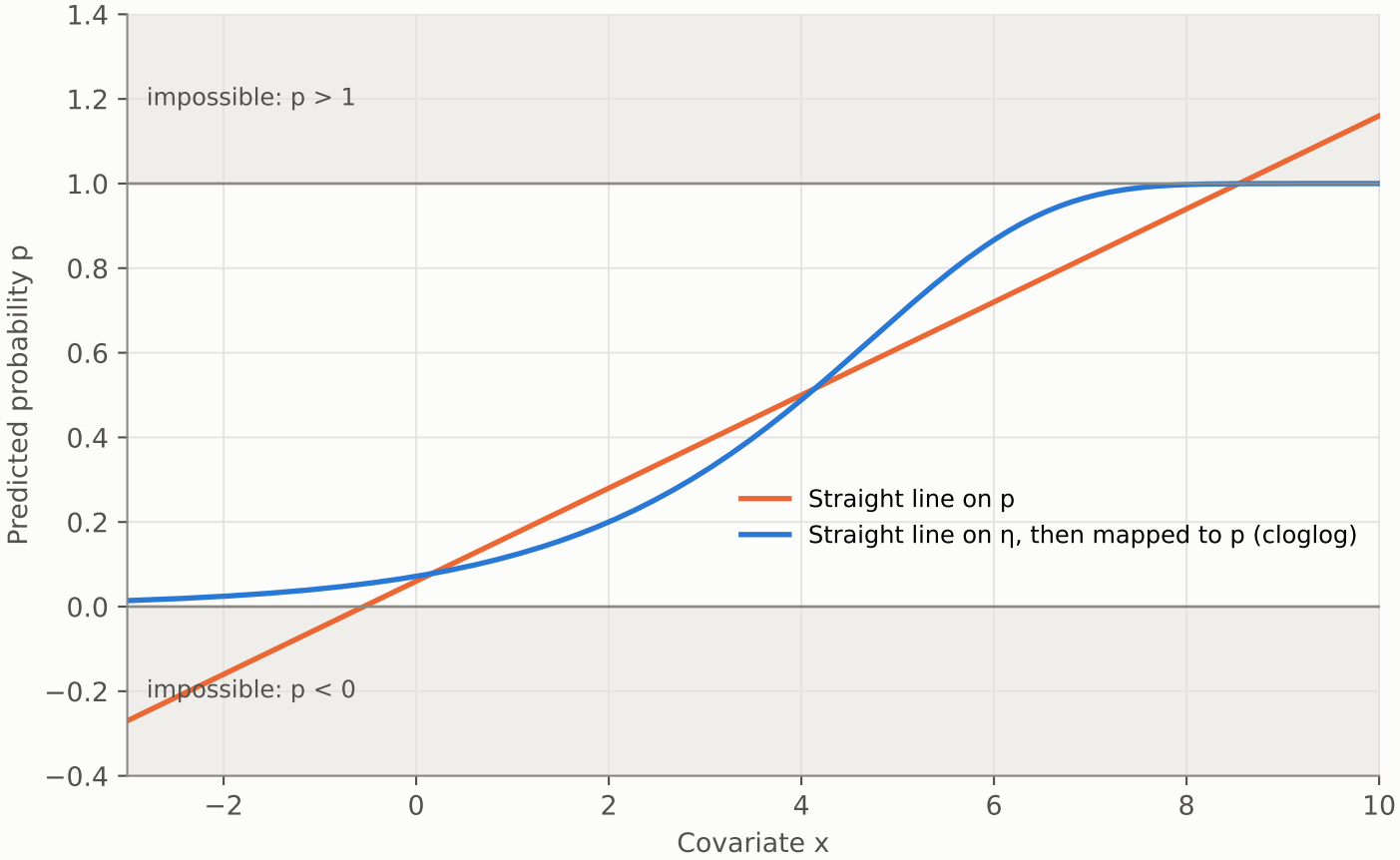


Read the figure from top to bottom. The six points are equally spaced on η, one unit apart. After the first step they are spread out to the right; after the second they are crowded together near 1. Equal steps on η are not equal steps on p. That bend is what "linear on a transformed scale" means: the straight line lives on η, and p is a curved function of it.

> **Each scale has its own job**
> η is where the model is fitted, because a weighted sum is simple to estimate and test. Λ is where the coefficients are read, because exp(β) is a hazard ratio. p is where results are reported, because it is the quantity a reader and the data understand. The next pages show why this split is a good idea.

Why not fit a straight line to p directly?

The obvious alternative is to put the weighted sum straight onto the probability, p = β0 + β1·x. A straight line has no walls, so for large or small enough x it leaves the interval from 0 to 1 and predicts probabilities that cannot exist.



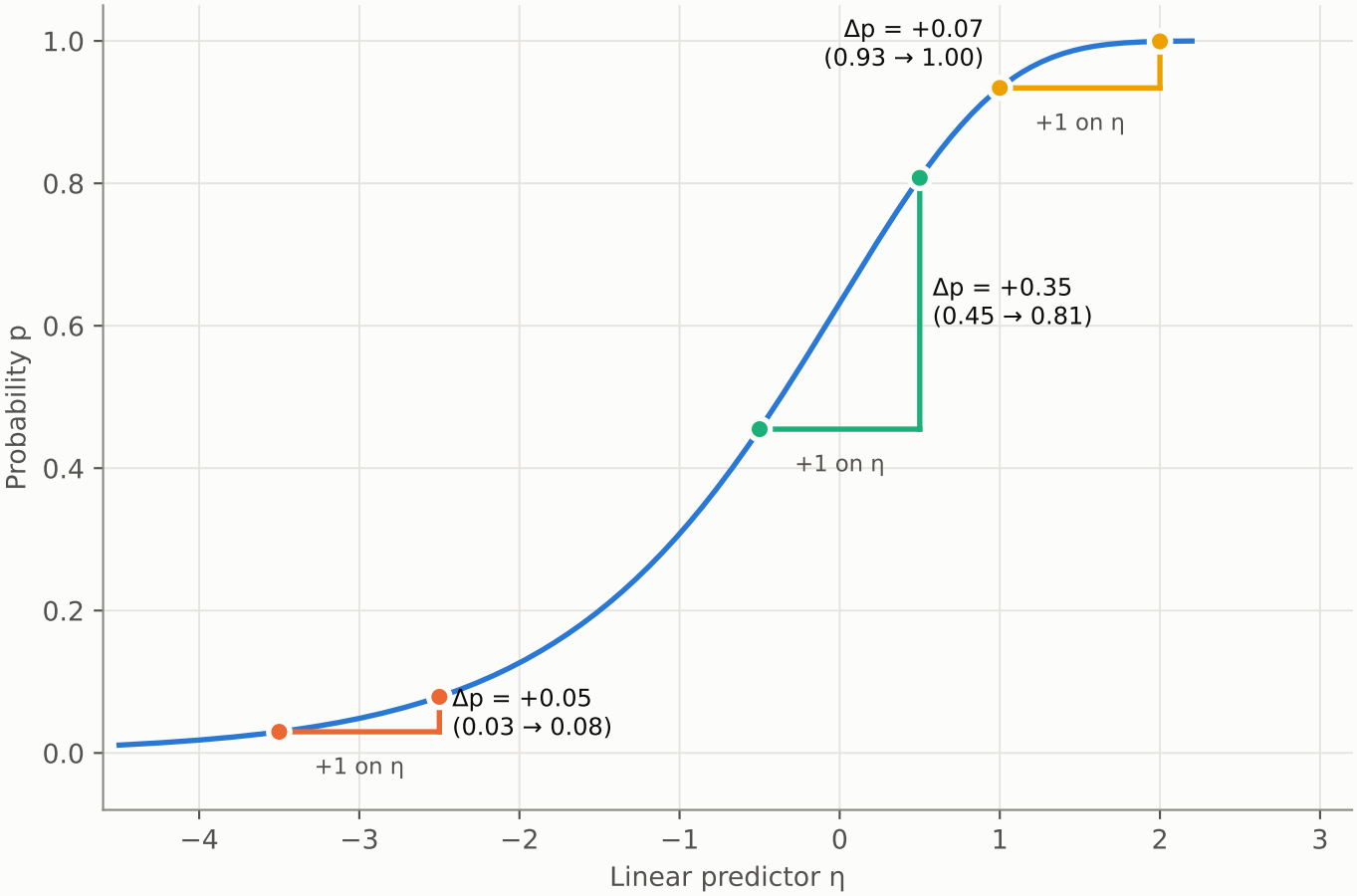
The blue curve is the same idea, a straight line in x, but drawn on η and then carried to p through the two steps from page 1. Whatever value η takes, the result stays between 0 and 1. The model can never predict an impossible probability, by construction and not by luck.

There is a second, quieter problem with the orange line. It says that one more unit of x always adds the same amount, here 0.11, to the probability. For a unit already at p = 0.95 that is impossible, and for a unit at p = 0.01 it would be an elevenfold increase. A constant additive effect on p does not make sense across the whole range of risk.

In one sentence
Probabilities are bounded and rates are not, so the model is made linear on a scale without bounds and only converted to a probability at the end.

One coefficient, many probability changes

Take a covariate with β = 1. Increasing it by one unit always moves η up by exactly 1, wherever the unit starts. On the probability scale the same step is small for a low-risk unit and large for a medium-risk unit, and small again near the top.



All three steps are the same effect, β = 1. On the hazard scale they are also the same: each multiplies the hazard by exp(1) ≈ 2.7. Only on the probability scale do they look different, because there the effect depends on the starting risk.

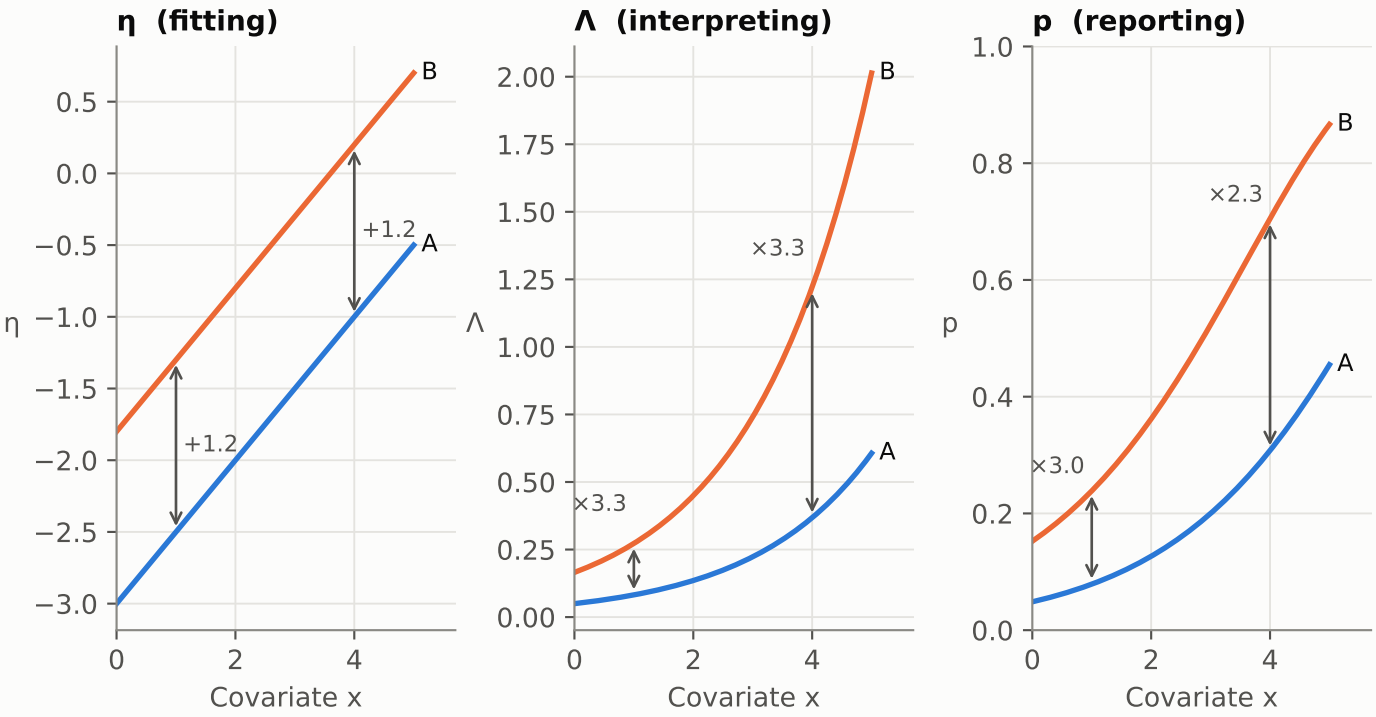
This is the practical advantage. One coefficient describes the covariate for every unit, whatever its baseline risk, and exp(β) is a hazard ratio that can be reported directly. On the probability scale a covariate would need a different number for every starting point.

Useful approximation for rare events
When risks are small, p ≈ Λ, because 1 − exp(−Λ) ≈ Λ for small Λ. The hazard ratio is then close to the ratio of probabilities (the relative risk). The orange step on the left shows this: the hazard is multiplied by 2.7 and the probability by about 2.6.

SEEING THE EFFECT FROM THREE ANGLES

The same two units on all three scales

Two groups of units differ only in one binary covariate with β = 1.2 (group B has it, group A does not). Both also share a continuous covariate x. The three panels plot the same model on the three scales.



Left: on η the two lines are parallel. The gap is 1.2 everywhere, which is exactly what "linear in the covariates" means. Effects add.

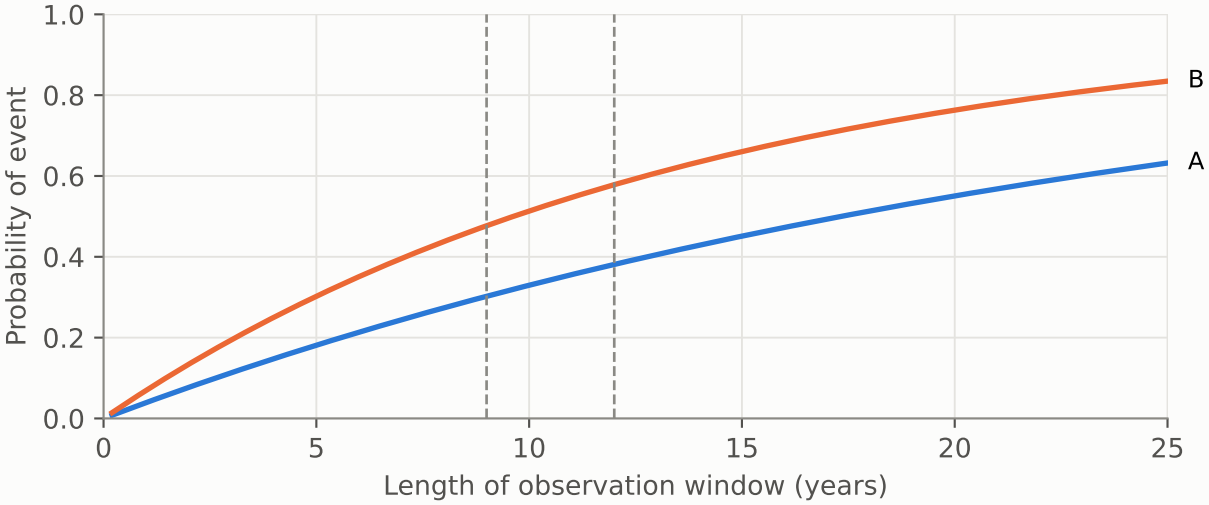
Middle: on the hazard scale the gap grows, but the ratio stays fixed: B always has exp(1.2) ≈ 3.3 times the hazard of A. Effects multiply, and the multiplier is the hazard ratio.

Right: on the probability scale neither the gap nor the ratio is constant. At x = 1 the probability in B is 3.0 times that in A; at x = 4 it is only 2.3 times, because B is approaching 1 and has less room left to grow.

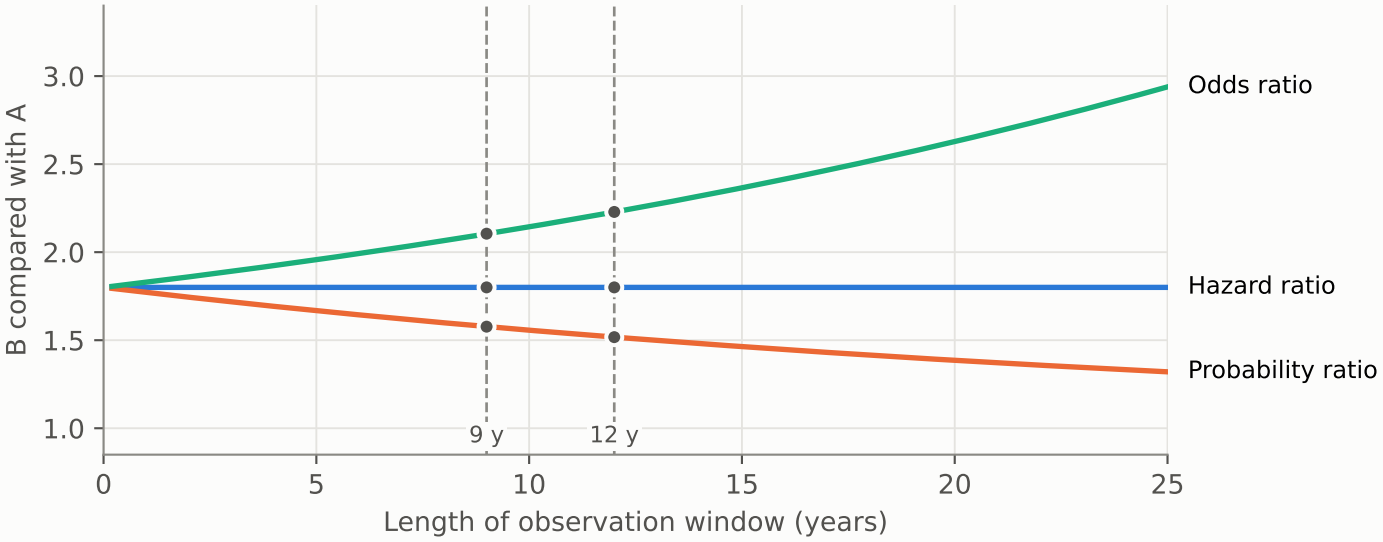
Why this is the smart arrangement
The model is fitted where effects add (η), read where effects multiply by a constant (Λ), and reported where the numbers mean something to a reader (p). Each scale is used for what it is good at, and the two fixed steps between them mean nothing is lost on the way.

Why 9 or 12 years of follow-up does not matter

Two units, A and B, have a constant hazard over time, and B's hazard is 1.8 times A's. The longer they are followed, the more likely each is to have had an event. The top panel shows their probabilities as the window grows; the dashed lines mark 9 and 12 years.



The bottom panel asks how much riskier B is than A, measured three ways. Only one of them gives the same answer at 9 and at 12 years.



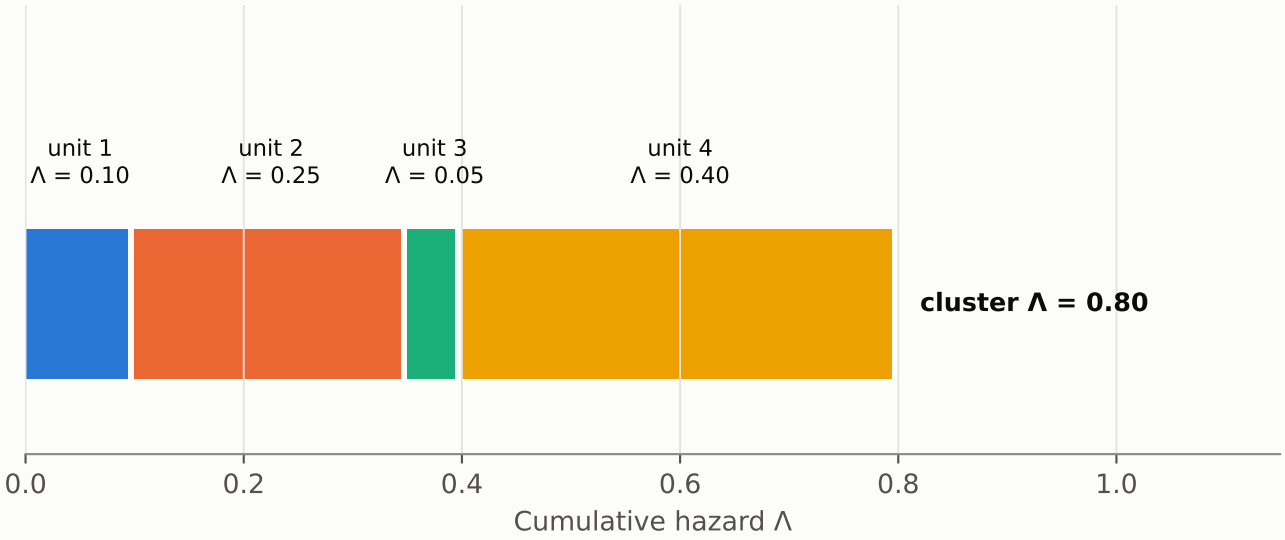
The hazard ratio is 1.8 whatever the window. The probability ratio falls (from 1.58 at 9 years to 1.52 at 12), and the odds ratio rises (from 2.10 to 2.23). If patients with different follow-up were pooled on the probability or odds scale, part of the estimated effect would really be a difference in follow-up time. On the hazard scale it is not.

How follow-up then enters the model

A longer window only adds risk that is the same for everyone: η moves up by log(window length). With an offset, a patient followed 12 years gets +log(12/9) ≈ 0.29 on η relative to one followed 9 years, and every β is left untouched. This holds if the hazard ratio really is constant over the window (proportional hazards); with one follow-up scan that assumption cannot be checked.

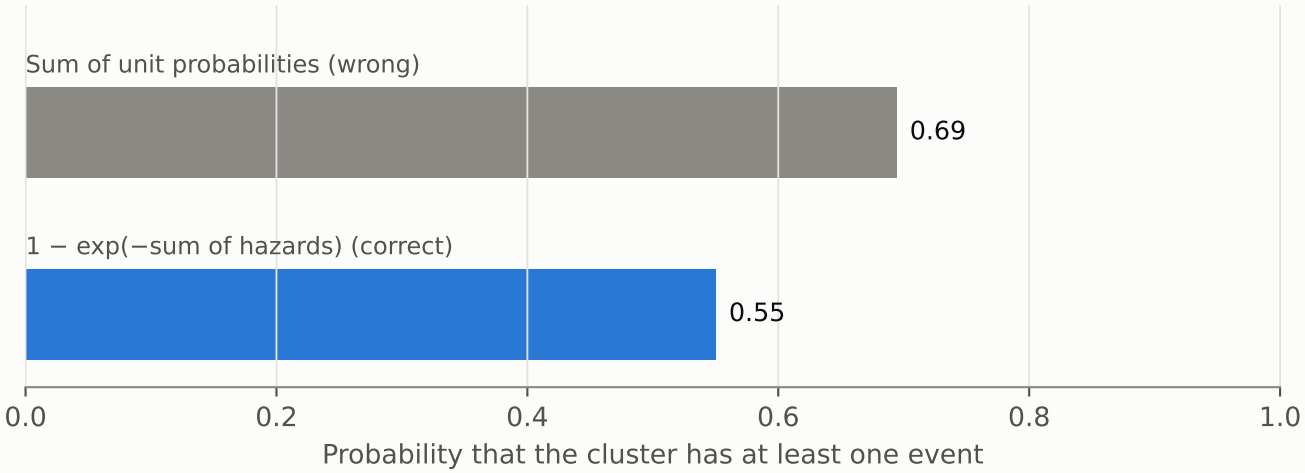
Adding up the units of a cluster

Often the question is not about one unit but about the whole cluster: does at least one of its units have an event? Here the hazard scale pays off a second time.



On the hazard scale the units simply stack: the cluster's cumulative hazard is the sum of its units' hazards. One more step gives the cluster probability, P(at least one event) = 1 − exp(−ΣΛ), which follows from multiplying the units' chances of staying event-free.

On the probability scale there is no such shortcut. Adding the unit probabilities double-counts the clusters where two units both have an event, so it overstates the risk.



With the four units above, adding probabilities gives 0.69, while the correct value is 0.55. The gap grows as unit risks get larger or clusters get bigger.

Link to the thesis text
This is the step in "From Unit Risk to Cluster Risk": given the cluster effect, the product of the units' event-free probabilities collapses to a single exponential of the summed hazards. It is exact only because the model is built on the hazard scale.

PUTTING IT TOGETHER

Summary

Scale	Range	One unit more of x	Used for
η (linear predictor)	$-\infty$ to ∞	adds β	fitting and testing
Λ (cumulative hazard)	0 to ∞	multiplies by $\exp(\beta)$	interpreting (hazard ratio)
p (probability)	0 to 1	depends on baseline risk	reporting and calibration

Why fitting on a transformed scale and reporting on the probability scale is smart:

1. Predictions can never fall outside 0 and 1, whatever the covariates (page 2).

2. One coefficient describes the effect for every unit, whatever its baseline risk (pages 3–4).

3. $\exp(\beta)$ is a hazard ratio that does not depend on the length of follow-up, so patients followed 9 and 12 years can be pooled (page 5).

4. Unit hazards add up to a cluster hazard, which gives the cluster risk exactly (page 6).

5. The final answer is still a probability, the scale on which risk is communicated and checked against outcomes.

The thesis sentence, in plain words
"Rearranging shows that the model is linear in the covariates on a transformed scale" means: the covariates are combined as a weighted sum, and that sum is set equal to $\log(-\log(1-p))$ rather than to p itself. The probability is recovered at the end by undoing the transformation.

What the scales do not fix
All of this rests on proportional hazards: the covariate multiplies the hazard by the same factor throughout the window. If that fails, the hazard ratio is an average over the window, and the window-length argument on page 5 weakens.